

UNIT: 05

NP Hard & NP Complete

P (Polynomial)

E (Exponential)

Linear search (n)

Binary search ($\log n$)

Insertion sort (n^2)

Merge sort ($n \log n$)

Matrix Multiplication (n^3)

0/1 Knapsack

TSP

8QS

Graph colouring

Hamilton cycle

2^n

The guidelines, rules to convert exponential TC to polynomial TC then it is called NP Hard and complete.

- 1) Similarity (similar products already exists.)
- 2) Relativity (how related it is to already existing)
- 3) D & ND

↓

↓

Deterministic
(already solved)

Non deterministic (research gap)
(left).

$N_{search}(A, n, Key)$

```

{
    j = choice(1); — ①
    if (Key == A[j])
    {

```

write(j);

} success(); — ①

write(0);

failure(); — ①

}

This is not done

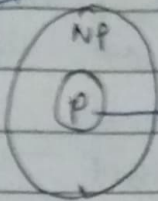
it is searching

time complexity

constant because this is

why it is non
deter.

After



$$P \subseteq NP$$

(Cook's theorem)

- Tools
- Notebook LM
 - Consensus
 - semantic scholar

either P is subset of NP or it is equal to NP -



Before

Satisfiability (already non dete. algo likha gaya hai).

$$(x_1 \vee x_2 \vee x_3) \wedge (\bar{x}_1 \vee x_2 \vee \bar{x}_3) \rightarrow \text{CNF}$$

$$(x_1 \wedge \bar{x}_2 \wedge x_3) \vee (\bar{x}_1 \wedge x_2 \wedge \bar{x}_3) \rightarrow \text{DNF}$$

Tautology $\neq$ Contradiction.

x_i : situations.

CNF : conjunctive normal forms

$\wedge \rightarrow$ conjunction / AND

DNF : disjunctive NF.

$\vee \rightarrow$ disjunction / OR.

x_1	x_2	x_3
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

we have to prove
for satisfiability.

$2^n \rightarrow$ possibilities.

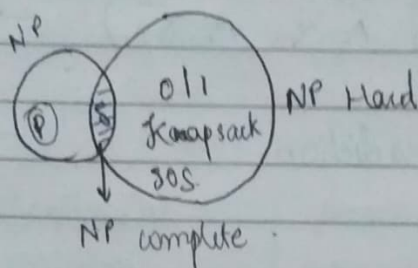
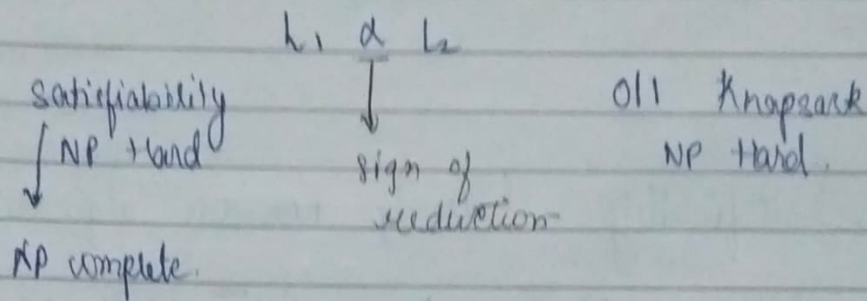
Here 2^3

Satisfiability problem $\rightarrow 2^n$

Es problems jiske basis pe doosri problems solve kar sake $\equiv$ NP hard.

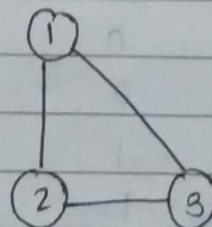
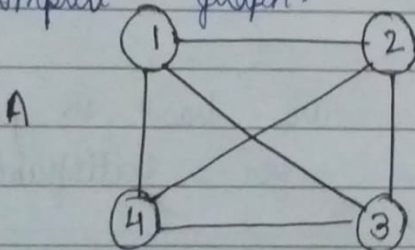
NP complete: All those NP hard problems for those already algorithms are written then it is NP complete.

All NP complete are then NP hard.



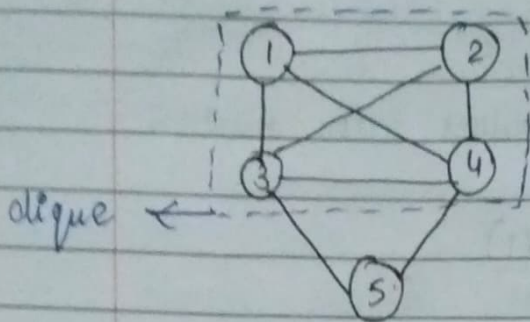
Problems which are NP Hard and NP both then it is NP complete like satisfiability.

complete graph:



$$|V| = n$$

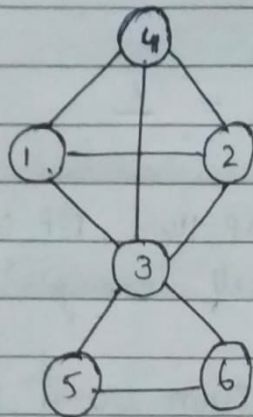
$$|E| = \frac{n(n-1)}{2}$$



Here graph is not complete but the subgraph is complete

The complete subgraph is known as clique

Clique Decision Problem :



$K=4$

$K=2$

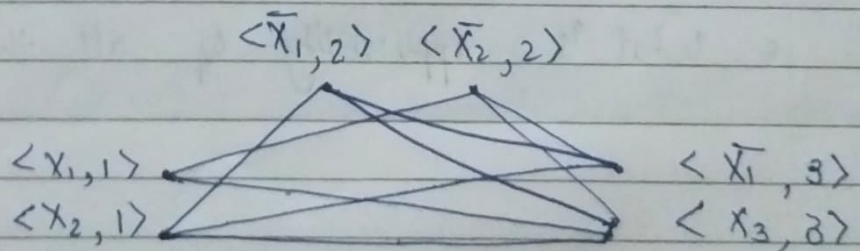
$K=3$

Decision : Is there any clique having $K=4$?
Optimization : Find out the clique having max K .

Ques

$$F = \underbrace{(x_1 \vee x_2)}_{C_1} \wedge \underbrace{(\bar{x}_1 \vee \bar{x}_2)}_{C_2} \wedge \underbrace{(\bar{x}_1 \vee x_3)}_{C_3}$$

$$F = \bigwedge_{i=1}^K C_i \quad F = \bigwedge_{i=1}^K C_i$$



Rule to connect :

$$(1) \quad E = \{ \langle a, i \rangle \langle b, j \rangle \mid i \neq j \text{ and } b \neq \bar{a} \}$$

Let X_1, X_2, X_3 put values in question
 $X=3$ 1 1 1

$$\frac{(0 \vee 1)}{C_1} \wedge \frac{(1 \vee 0)}{C_2} \wedge \frac{(1 \vee 1)}{C_3}$$

$$C_1 = 1$$

$$C_2 = 1$$

$$C_3 = 1$$

$$\Rightarrow C_1 \wedge C_2 \wedge C_3$$

$$\Rightarrow 1 \wedge 1 \wedge 1 = 1$$

Assignment 1: Explain P, NP, NP Hard, NP complete problems with examples (self examples).

2. Differences b/w decision and optimization problem.

3. Explain CDP.

4. Explain satisfiability.

5. Explain reducibility and its properties.

6. What is applicability of all these problems.

* String Matching

KMP

→ Pattern matching

prefix = suffix

Knuth Morris Pratt

pattern $\Rightarrow$ abcabc

prefix : a, ab, abc, abcd

suffix : c, bc, abc, dabc

Ex: $\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ a & b & c & d & e & a & b & c & d & e \end{matrix}$
 $\rightarrow$ π table

string: $\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 \\ a & b & a & b & c & a & b & c & a & b & a & b & a & b & d \end{matrix}$
 $\begin{matrix} 0 & 0 & 1 & 2 & 0 & 1 & 2 & 5 & 1 & 2 & 1 & 2 & 1 & 2 & 0 \end{matrix}$

Time complexity

$O(m+n)$

π table to search an alphabet.

* Rubin Karp :

a=1

b=2

c=3

d=4

e=5

f=6

g=7

h=8

i=9

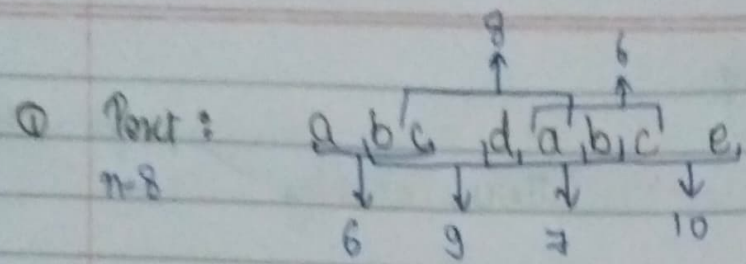
j=10

Text

$\begin{matrix} & & 3 & & 4 \\ & & \uparrow & & \uparrow \\ a & a & a & a & b \\ \downarrow & & \downarrow & & \\ 3 & & 3 & & \end{matrix}$

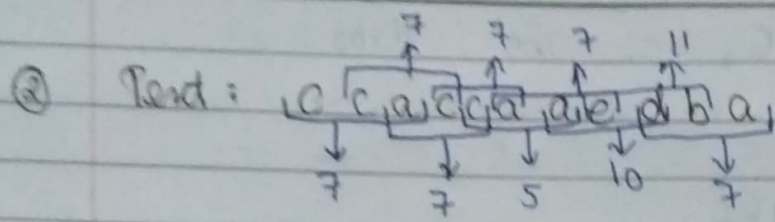
pattern = aab

$1+1+2=4$



pattern: aab
 $1+1+2=4$

no 4 no match
 pattern not found.



pattern: $d b a$
 $4+2+1=7$

- $(3 \times 10^2 + 3 \times 10^1 + 1 \times 10^0)cca : 7 \rightarrow 331$
 $300 + 10 + 3 \quad cae : 7 \rightarrow 313$
 $100 + 30 + 3 \quad acc : 7 \rightarrow 133$
 $cca : 7 \rightarrow 331$
 $caa : 5 \rightarrow 311$
 $cae : 7 \rightarrow 115$
 $aed : 10 \rightarrow 154$
 $edb : 11 \rightarrow 542$
 $dba : 7 \rightarrow 421$

$d \ b \ a$
 $4 \ 2 \ 1$
 $4 \times 10^2 + 2 \times 10^1 + 1 \times 10^0$
 421

3 char & power.

{ matched }
 Pattern exists.

spurious

hit / collision

Time complexity: $\Theta(n-m+1)$

cca
 cae

$$[(3 \times 10^2 + 3 \times 10^1 + 1 \times 10^0) - 3 \times 10^2] \times 10 + 3 \times 10^0$$

$$\uparrow$$

$$\text{pattern length}$$

$$\Theta(n-m+1)$$

$$\uparrow$$

$$\text{text length}$$